

Alberta opioid mortality around the COVID-19 onset

An interrupted time series analysis using federal data

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github.com/Adilbek123/alberta-substance-use-trends

COVID-19 hit Alberta in March 2020, and many other things shifted at the same time

- COVID-19 was declared a pandemic by the WHO on 11 March 2020
 - Alberta declared a public health emergency on 17 March 2020
 - Roughly 60,000 Canadians have died from COVID-19 directly between 2020 and 2023
- Beyond the virus itself, the pandemic disrupted multiple systems at once
 - Drug supply chains for the unregulated drug market
 - Harm reduction services (e.g., supervised consumption sites operating at reduced capacity)
 - Mental health and addiction services (in-person care disruption, telehealth ramp-up lag)
 - Employment, housing, and social connection
- Other provinces reported a surge in opioid deaths from 2020 onward
 - British Columbia, Ontario, and others saw sharp increases at roughly the same time
 - Alberta's pattern has not been quantified at this granularity in public analysis

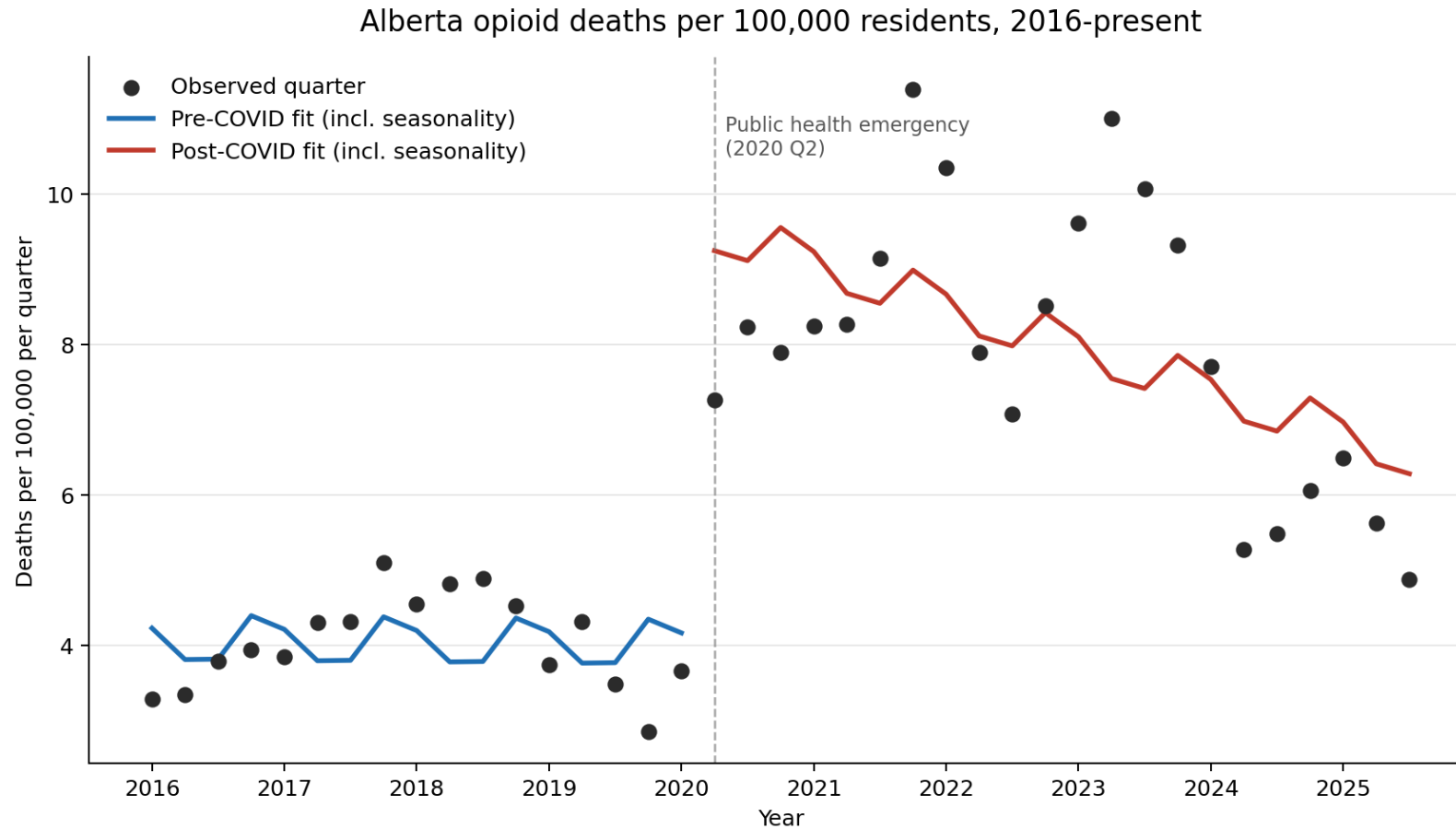
Did Alberta opioid mortality jump at the COVID onset, and is it real or noise?

- Pre-COVID context
 - Alberta opioid death rate sat around 4 per 100,000 residents per quarter, 2016 through early 2020
 - Flat trend across that period, no sustained rise or fall
- If a jump occurred at COVID, two things matter for policy
 - How large was it? This sets the post-COVID baseline against which Recovery Model progress will be measured
 - Is it real, or could random variation in a noisy series explain it? This determines whether the change calls for action
- This analysis answers both questions
 - Whether the level shift exists at the cutoff, separately from the existing trend
 - Whether the size of the shift is meaningfully larger than what random variation could produce

An interrupted time series compares the pre-COVID and post-COVID periods statistically

- **Method: interrupted time series (also called segmented regression)**
 - Fits one trend to the pre-COVID years, another to the post-COVID years
 - Measures the gap between them at the cutoff (2020 Q2, first full quarter under the emergency)
 - Tests whether that gap is meaningfully larger than the natural noise in the data
- **Data**
 - Opioid deaths: Public Health Agency of Canada Health Infobase, Alberta quarterly, 2016 Q1 to 2025 Q3
 - Population: Statistics Canada Table 17-10-0009-01, used to convert counts to a rate per 100,000
 - No internal Government of Alberta data is used; all sources are public
- **Robustness checks confirm the result is not an artifact of method choices**
 - Placebo cutoffs placed in the pre-period (should produce no jump if the design is credible)
 - Drop the quarter that straddles the cutoff and refit
 - Cross-check using a count model with a population offset

The death rate roughly doubled at 2020 Q2, sharp, large, and robust



- Pre-COVID average
 - About 4 deaths per 100,000 per quarter
- Post-COVID average
 - About 8 deaths per 100,000 per quarter
- Level shift at the cutoff
 - About 5.5 per 100,000 per quarter
 - Roughly a doubling
 - Statistically very strong ($p < 0.001$)
- Pre-COVID trend was flat
 - The jump is not part of an existing rise
- All robustness checks hold
 - Placebos produce no jump
 - Drop-quarter estimate moves $< 3\%$
 - Cross-check confirms

Sources: PHAC Health Infobase substance-related harms data; StatCan Table 17-10-0009-01.

Level shift at 2020 Q2 (segmented regression with quarter FE, HAC SE 3-lag, small-sample correction): +5.50 per 100k [95% CI +3.60, +7.39].

The pandemic as a whole shifted opioid mortality, not the virus alone

- The total effect attributable to the pandemic, including its downstream consequences
 - Virus, public health measures, drug supply disruption, harm reduction capacity, mental health service disruption, isolation, economic shock
 - These came together by construction, not by coincidence; a pandemic without them is not a counterfactual that exists in any data
- The analysis cannot separate which channel did the most damage
 - Decomposing into supply vs. services vs. isolation vs. economic shock requires different data
 - Drug-checking composition, service utilization series, and similar mechanism-level sources
- Comparing Alberta to other provinces would not isolate a virus-only effect
 - Every Canadian province faced COVID and its policy response at essentially the same time
 - A peer comparison would identify Alberta-specific deviation from the common pattern, not the virus channel alone

Post-COVID baseline is higher than pre-COVID, and recovery is already underway

- For Recovery Model performance measurement, the baseline matters
 - The post-COVID baseline is roughly double the pre-COVID baseline
 - Any 'return to baseline' framing needs to be explicit about which baseline
- The trajectory is not flat: a real recovery is already visible
 - The post-COVID period peaked in 2021 Q4 and has been declining since 2024
 - A single average over the post-period masks this dynamic
- The next analytical work is decomposition, not better identification
 - Which channel drove most of the shift (supply, services, isolation, economic)
 - Which age groups, geographic zones, and substance types concentrate the increase
 - Modelling the rise-to-peak-and-decline shape directly, rather than a single level shift
 - Cumulative excess deaths through end-of-sample as a policy-relevant total